

Economists call the level curves of a utility function *indifference curves* and those of a production function *isoquants*. Thus, level curves have algebraic equations of the form $f(x, y) = k$ for all possible values of k . A contour diagram or contour map of a function $f(x, y)$ is a 2-dimensional graph (Fig. 4.43) showing several level curves in the xy -plane corresponding to several values of k .

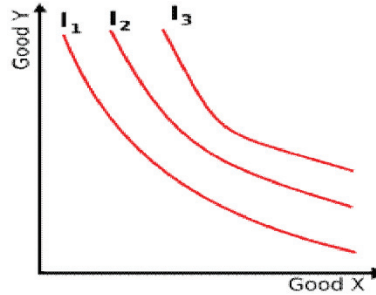


Fig. 4.43: Indifference Curve Map

Example: Cobb-Douglas Production Model

Given P = the monetary value of all goods produced, K = total capital investment over the same period, and L = the total labor force, the following relationship between these quantities holds: $P = 1.01L^{0.75}K^{0.25}$

The contour diagram shown in Fig. 4.44 below shows that reducing labour

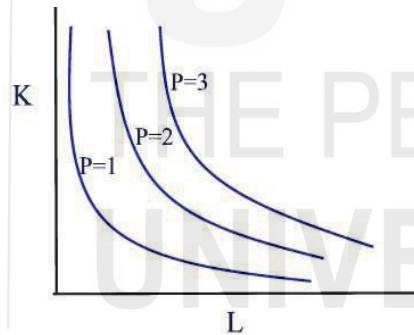


Fig. 4.44: Indifference Curve Map- Cobb Douglas Example

increases capital investment in order to maintain the same level of production i.e., supposedly more investment is needed in machinery to make up for the lack of available labour. Conversely, increasing labour reduces capital investment since supposedly the work is being done by the labor force.

Example:

Let $f(x, y) = x^2 - y^2$. We will study the level curves $c = x^2 - y^2$.

First, look at the case $c = 0$. The level curve equation $x^2 - y^2 = 0$ factors to $(x - y)(x + y) = 0$. This equation is satisfied if either $y = x$ or $y = -x$. Both these are equations for lines, so the level curve for $c = 0$ is two lines.

If $c \neq 0$, then we can rewrite the level curve equation $c = x^2 - y^2$ as $1 = x^2/c - y^2/c$.

See that this is the equation for a hyperbola. If c is positive, the hyperbolas open to the left and right. If c is negative, the hyperbolas open up and down.

For example, if $c=1$, the equation is $x^2-y^2=1$. If $c=-1$, the equation is $x^2-y^2=-1$. Several level curves are plotted in Fig. 4.45.

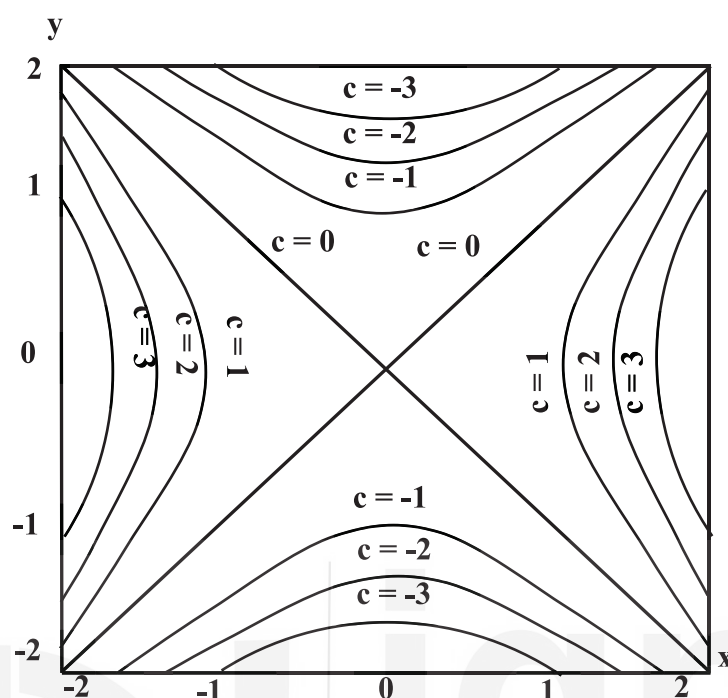


Fig.4.45: Level Curves

Check Your Progress 7

- 1) Explain the meaning of level curve.

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- 2) Let $f(x,y) = xy$ for all (x,y) . Show a collection of level curves of f .

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- 3) Suppose a function is defined by $f(x,y)=1$ for all (x,y) . Comment on the shape of level curves.

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4.9 LET US SUM UP

In this unit we discussed mainly the method of sketching graphs of functions. However, first we discussed how to obtain algebraic equations corresponding to some well-known geometric figures. In the reverse direction, for the purpose of graphing functions, starting with relationship between elements of domain and co-domain, we recalled the basis of functional relationships. An operational method of dealing with a function was pointed out to identify dependent and independent variables. Thus, we discussed $y = f(x)$, it would be to treat x as the independent variable picked up from among the elements of domain and y to be the dependent variable present as an element in the range. Such a formulation helped translate the functional relationships into graphical representation in the rectangular co-ordinate system where x -axis would contain the values taken by independent variables and y -axis is the values of dependent variables. Before proceeding with a process of sketching graph of a function, the use of vertical line test for testing a relationship for its possibly being a functional relationship was discussed. According to this test, equations of circle and ellipse etc. fail to meet its requirement of functional relationship.

While dealing with the method of drawing the graph, the unit considered functions in five groups, viz., linear, curve, asymptotic, piece-wise and curve with branches. The linear types of graphs were drawn when the functions would be in the nature of linear, absolute value or step-like. In cases of graphs showing curves, functions included were odd and even, quadratic and cubic. Among the graph of asymptotic types, functions included were square root, exponential and logarithmic. Graphs appearing in pieces in the xy -plane comprised rational functions, piecewise functions and discontinuous functions. In the last group of graph shaving *curve* with branches functions concerned with hyperbola and rectangular hyperbola were discussed. In the last part of the present unit, graphs of functions with two variables were introduced.

4.10 KEY WORDS

Abscissa	: x -coordinate (horizontal) of a point in a 2-dimensional coordinate system.
Absolute Value Function	: Obtaining the <i>absolute value</i> (or <i>magnitude</i>) of a number.
Asymptote	: A line that a curve approaches, as it heads towards infinity.
Asymptotic Discontinuity	: Graph approaching a point but never touching the point.
Axis of Symmetry	: A line of symmetry for a graph. The two sides of a graph on either side of the axis of symmetry look like mirror images of each other.

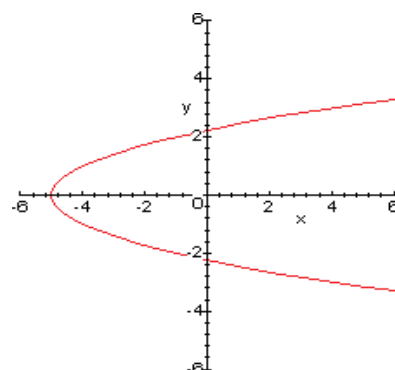
Cobb-Douglas Production Function	: Production function which is applied to describe how much output two inputs into a production process make.
Contour Diagram (contour map)	: Graph that provides one way of visualizing functions of two variables.
Co-ordinates	: Any of a set of two or more numbers used to determine the position of a point, line, curve, or plane in a space of a given dimension with respect to a system of lines or other fixed references.
Cubic Function	: A polynomial function of degree 3.
Directrix	: A line perpendicular to the axis of symmetry of a parabola.
Discontinuous Function	: A function which for certain values or between certain values of the variable does not vary continuously as the variable increases.
Even Function	: Univariate function is $f(x)$ is said to be even provided that $f(x)=f(-x)$.
Exponential Function	: Function defined by $\exp(z)=e^z$.
Hyperbola	: A hyperbola is a curve where the distances of any point from: a fixed point (the focus), and a fixed straight line (the directrix) are always in the same ratio.
Indifference Curve Map	: A group of <i>indifference curves</i> for two commodities showing different levels of satisfaction.
Isoquants	: A graph of all possible combinations of inputs that result in the production of a given level of output.
Jump Discontinuity	: A function that jumps from one location to another at a certain point.
Level Curves	: A contour diagram or contour map of a function $f(x,y)$ is a 2 dimensional graph showing several level curves in the xy -plane.
Logarithmic Function	: A function (as $y = \log_a x$ or $y = \ln x$) that is the inverse of an exponential function (as $y = a^x$ or $y = e^x$) so that the independent variable appears in a logarithm.
Odd Function	: Univariate function $f(x)$ is said to be odd provided that $f(-x)=-f(x)$.

Ordinate	:	Element of an ordered pair which is plotted on the vertical axis of a two-dimensional Cartesian coordinate system.
Piecewise Function (split function)	:	A function can be in Pieces.
Point Discontinuity	:	Graph of the function has a hole at $x=a$.
Polar Coordinates	:	A two-dimensional coordinate system in which each point P on a plane is determined by a distance r from a fixed point O that is called the pole (or origin) and an angle θ from a fixed direction.
Quadratic Function	:	A function that can be described by an equation of the form $(x)=ax^2+bx+c$, where $a \neq 0$.
Rational Function	:	A function that is the ratio of two polynomials.
Rectangular Co-ordinate System	:	Two real number lines that intersect at a right angle.
Rectangular Hyperbola	:	A hyperbola for which the asymptotes are perpendicular.
Square Root Function	:	A function that maps these to f non-negative real numbers onto itself.
Step Function	:	A function whose graph looks like a series of steps.
Vertex	:	A synonym for anode of a graph.
Vertical Line Test of Function	:	A vertical line anywhere on a graph so that it hits the graph in more than one spot, then the graph is not a function.

4.11 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISE

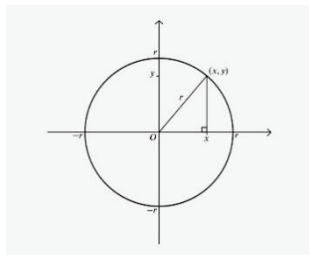
Check Your Progress 1

1) The graph is



By the vertical line test, this graph is not the graph of a function, because there are many vertical lines that hit more than once.

- 2) As in the case of straight line, for translating the geometric figure a circle, create a coordinate system around the object, with centre of circle as origin of coordinate system, as shown below.



Mark an arbitrary point, say P, on the circle, as (x, y). Draw a perpendicular from P onto X-axis, meeting it in say point Q. Then length of perpendicular is of length y. and x is the length of the intercept on X-axis. Then from conventional/ synthetic geometry, we know

$$x^2 + y^2 = r^2$$

is an algebraic equation of the circle with radius r.

Check Your Progress 2

- 1) First prepare the t-chart which is Table :t-chart of $y = (-5/3)x - 2$

x	$y = (-5/3)x - 2$
-6	8
-3	-2
0	2
3	-7

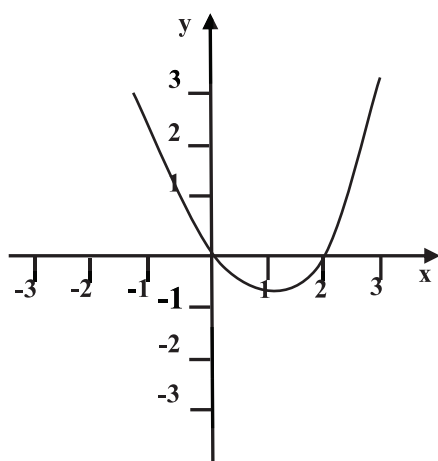
Then you draw the graph.

- 2) $x = 4/3$ or $x = -8/3$

$$3) f(x) = \begin{cases} 25 + 0.30x, & \text{for } 0 < x \leq 500 \\ 100 + 0.15x, & \text{for } 500 < x < 1000 \end{cases}$$

Check Your Progress3

- 1)



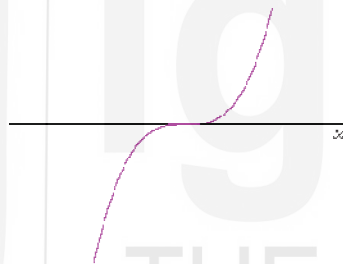
- 2) A curve like 's'.

Check Your Progress 4

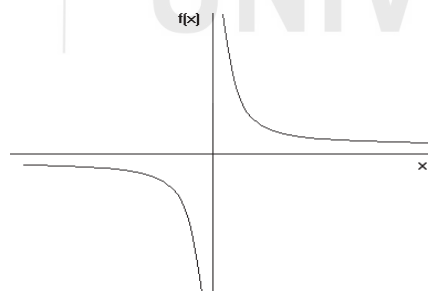
- 1) The x intercept is found by solving $f(x)=0$ or, $x+1=0$. The x intercept is at the point $(-1, 0)$. The y intercept is at the point $(0, f(0)) = (0, -1)$.
- 2) Since total fixed cost is a constant quantity, average fixed cost will steadily fall as output increases without becoming zero.

Check Your Progress 5

- 1) The function can be written $f(x)=ax^2+c$, where $a=-1$ and $c=4$. Since a is -1 , the graph is reflected across the x -axis. Since $c=4$ the graph is translated up 4 units. So, the graph of $y=-x^2+4$ is the graph of $y=x^2$ reflected across the x -axis and translated up 4 units. Now you draw the graph.
- 2) The cubic function is $y=x^3$



- 3) $f(x) = 1/x$ is a discontinuous function. Either it has a value at which $f(x)$ does not exist or has an asymptote. It is discontinuous at $x=0$ because $f(x)$ is undefined there. The discontinuity is visible in the graph.

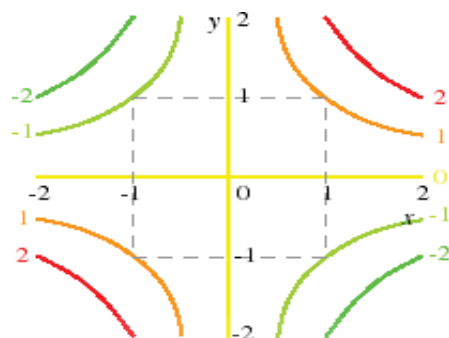


Check Your Progress 6

- 1) The rectangular hyperbola in general is defined as a hyperbola which has orthogonal asymptotic lines.
- 2) Price elasticity equal to unity.
- 3) Hint: write the standard form of a hyperbola with horizontal transverse axis.

Check Your Progress 7

- 1) A curve in two dimensions on which the value of a function $f(x, y)$ is a constant.
- 2) The following figure may be seen. The number beside each curve is the value of the function to which it corresponds. The level curve for the value 0 consists of the axes.



A level curve is defined as a *set*, and indeed some level curves of some functions are not "curves" at all. In the given function the value of f is 1 for all values of x and y . As a result you get straight lines, not curves. Moreover, the level curve of this function for the value 2 is empty as there are *no* values of (x, y) such that $f(x, y) = 2$.

4.12 EXERCISES

- Q1.** Find the vertex and intercepts of $y = 3x^2 + x - 2$ and graph; identify the vertex and the axis of symmetry.

Ans. To find the y -intercept, set x equal to zero and solve:

$$y = 3(0)^2 + (0) - 2 = 0 + 0 - 2 = -2$$

Then the y -intercept is the point $(0, -2)$. To find the x -intercept, set y

Equal to zero, and solve:

$$0 = 3x^2 + x - 2$$

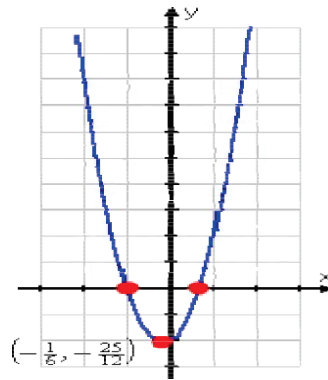
$$0 = (3x - 2)(x + 1)$$

$$3x - 2 = 0 \text{ or } x + 1 = 0$$

$$x = 2/3 \text{ or } x = -1$$

From the above, the x -intercepts are at the points $(-1, 0)$ and $(2/3, 0)$. The axis of symmetry is half way between the two x -intercepts at $(-1, 0)$ and at $(2/3, 0)$.

The complete answer is a listing of the vertex, the axis of symmetry, and all three intercepts, along with an ice, neat graph:

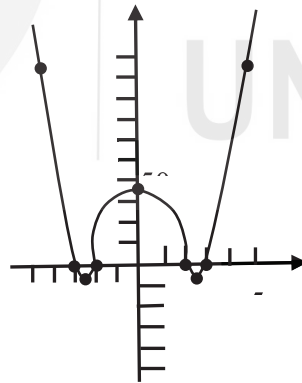


The vertex is at $(-\frac{1}{6}, -\frac{25}{12})$, the axis of symmetry is the line $x = -\frac{1}{6}$, and the intercepts are at $(0, -2)$, $(-1, 0)$, and $(\frac{2}{3}, 0)$.

Q2. Let $y = x^4 - 13x^2 + 36$. Draw its graph.

Ans. This is a positive even power ("to the fourth"), so the graph will be upon both ends. Factoring the polynomial, we get $y = (x+3)(x-3)(x+2)(x-2)$, so the zeroes (x-intercepts) are -3 , -2 , 2 , and 3 . If you plot a few other points on your T-chart, it will be no trouble to graph this:

x	$y = x^4 - 13x^2 + 36$
-4	84
-2.5	-6.19
-1	24
0	36
1	24
2.5	-6.19
4	84



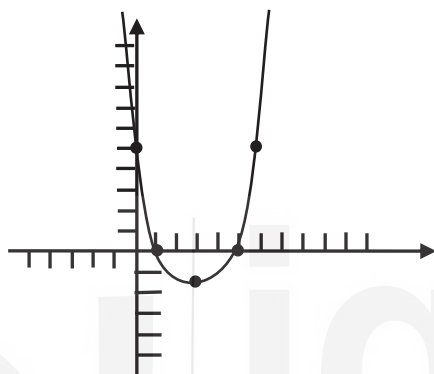
Graph of $y = x^4 - 13x^2 + 36$

Q3. A quadratic function is given as $y = x^2 - 6x + 5$. Draw its graph.

Ans. We need to plot more than three points to get the correct shape. Find the intercepts, which in this case are at $(1, 0)$, $(5, 0)$, and $(0, 5)$; find the vertex, which in this case is at $(3, -4)$. Remember that parabolas look like "smilies", not like straight lines. So, we want to plot a few more points:

Table: Chart of $y=x^2+6x+5$

x	$y = x^2+6x+5$
0	5
1	0
3	-4
5	0
6	5



Graph of $y=x^2+6x+5$

Q4. Sketch the graph of $f(x)=(x-1)^3+3$

Ans. We need to pick values of x to plug-in. Here are the function evaluations.

x	$f(x)$	(x,y)
-1	-7	$(-1, -7)$
0	0	$(0,0)$
1	1	$(1,1)$
2	2	$(2,2)$
3	9	$(3,9)$

You can sketch its graph now.

Q5. Sketch the graph of the following piece wise function.

$$g(x)= \begin{cases} -x^2+4 & \text{if } x < 1 \\ 2x & \text{if } x > 1 \end{cases}$$

Ans. Recall that graphing piecewise functions are like graphing several functions at once, except we are only going to graph them on very specific intervals. So, we will graph the following two functions,